

Understanding and Managing White Spots on Anodized Aluminum

July 25, 2019



Agenda

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- What is White Spotting?
- Factors that Contribute to White Spotting/Staining
- Surface Defect Sources
- Methods to Mitigate Surface Defects
- Q&A

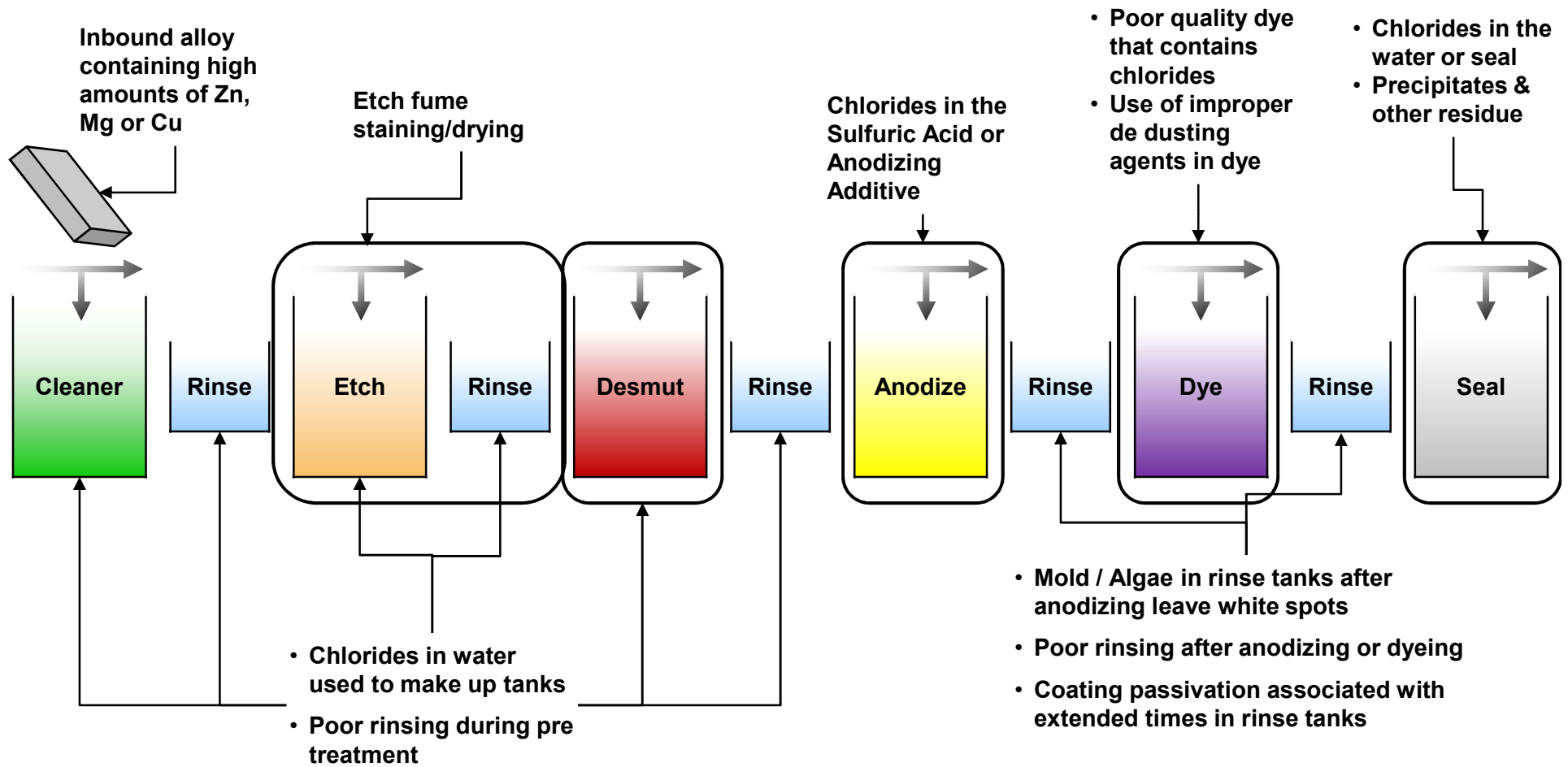
White Spotting: Understanding and Overview

- White Spotting/Staining is a common phenomenon, however understanding the cause of these surface defects can be challenging.
- The primary source for this confusion is that white spotting is not caused by the same issue all the time and in many cases, it could be a combination of multiple factors that result in the surface defect.
- Furthermore some defects are related to the actual anodic coating whereas others could simply be on the surface of the coating.
- In the list below we have summarized the key properties of the anodizing process that can lead to white spots or other surface defects:
 - ***Alloy Type***
 - ***Machining Quality / Part Profile***
 - ***Pretreatment process***
 - ***Orientation/ Airflow on production line***
 - ***Anodizing Conditions***
 - ***Coating Passivation***
 - ***Rinse Cycle***
 - ***Dye Conditions***
 - ***Sealing process***

White Spotting & Surface Defects: Understanding and Overview

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The graphic below shows the multitude of different process tanks and areas that could result in surface defects that look like white spots



Contributing Factors: Alloy type

Galvanic Activity



2000

No Galvanic Activity

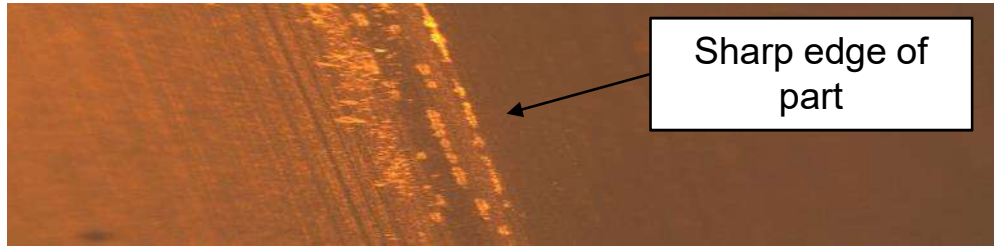


6000

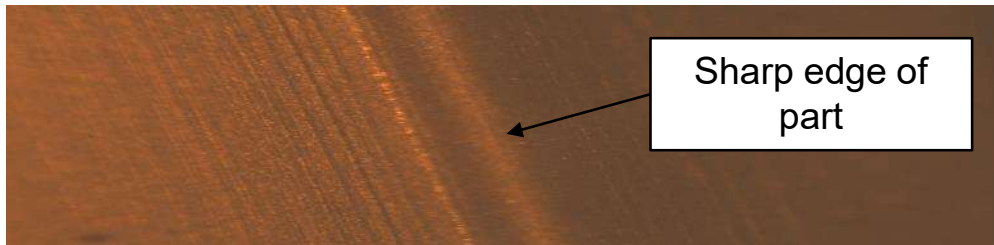
- Sources of spotting/staining
 - Certain alloy constituents are more susceptible to galvanic activity than others (2000, 7000)
 - Alloys with high levels of Cu, Mg & Zn
 - Rinse water corrosion (Chlorides and pH)
 - Chlorides in Dye tank
- Potential Solutions
 - Understand the alloy type
 - Manage rinsing – Reduce Immersion time, improve water quality
 - Use sacrificial anodes (Magnesium bars)
 - Chloride management – determine source and contain
 - Review tooling (Al vs Ti)

Contributing Factors: Machining Quality / Part Profile

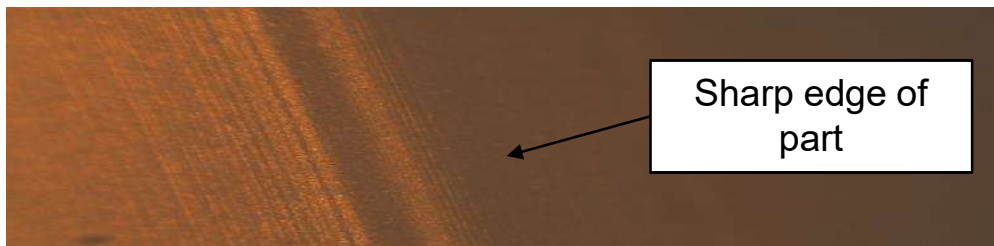
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High Chloride level



Moderate Chloride level

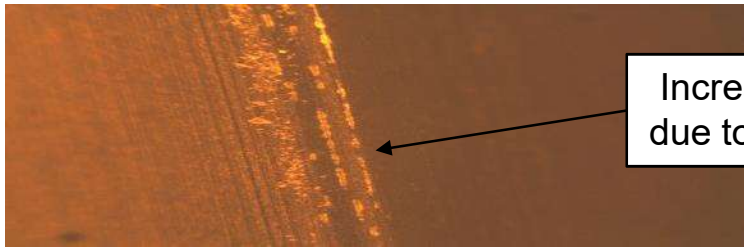


Low Chloride level

- Sources of spotting/staining
 - Sharp edges on parts
 - Contaminated cutting fluid
 - Poor material handling procedures
- Potential solutions
 - Check parts for sharp edges
 - Test cutting fluids for algae/mold that can cause corrosion
 - Use sacrificial anodes (Magnesium)
 - Chloride management – source and contaminant
 - Review tooling (Al vs Ti)

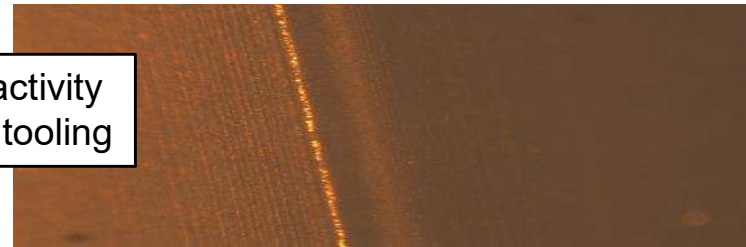
Contributing Factors: Machining Quality / Part Profile (Effect of Tooling)

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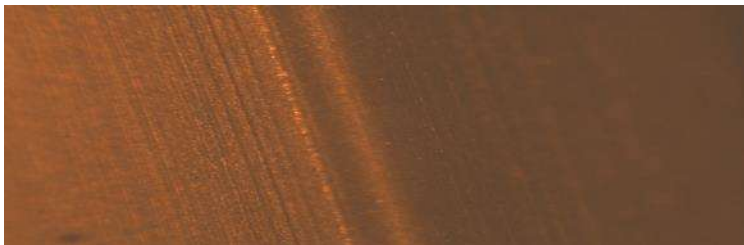


Increased galvanic activity
due to sharp edge & tooling

High Chloride level Ti Tooling



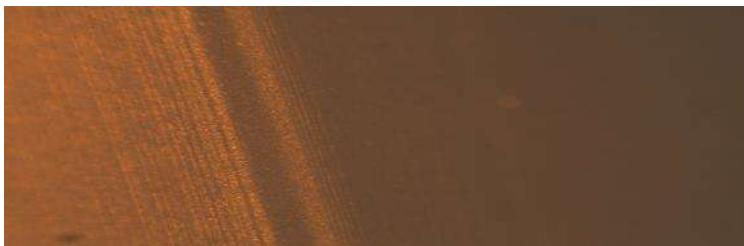
High Chloride level Al Tooling



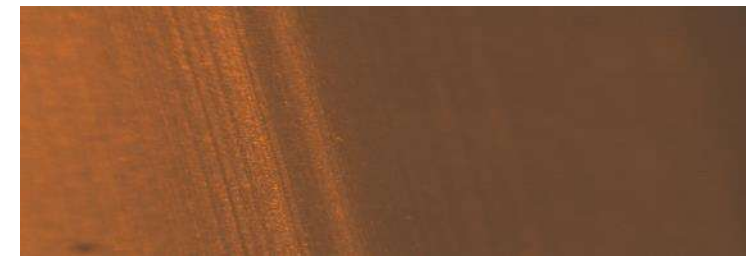
Moderate Chloride level Ti Tooling



Moderate Chloride level Al Tooling

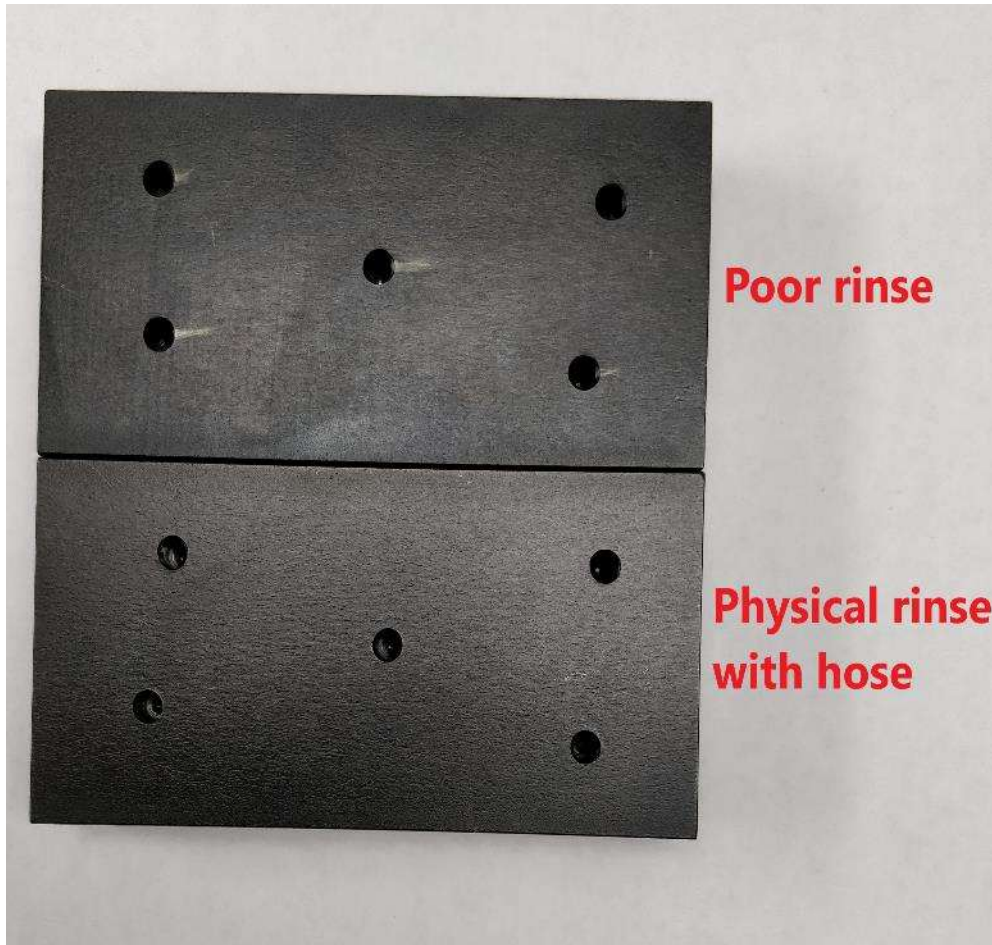


Low Chloride level Ti Tooling



Low Chloride level Al Tooling

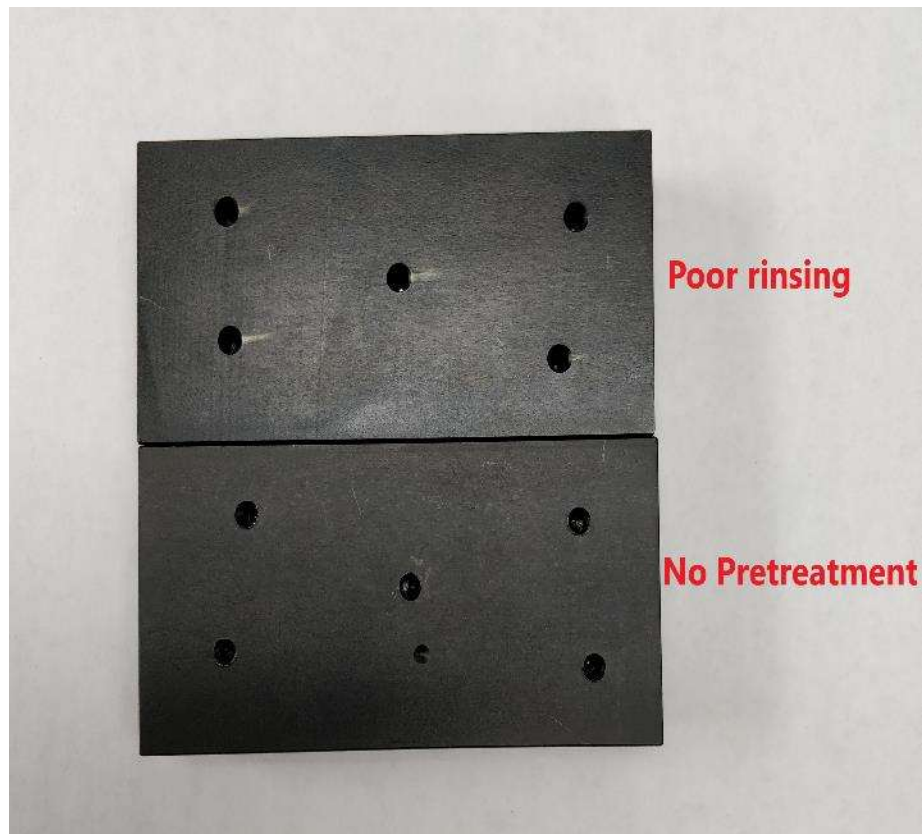
Contributing Factors: Pretreatment process



- Sources of spotting/staining
 - Threaded blind holes
 - Smaller the hole the greater the problem
 - Poor rinsing
 - Entrapment of process chemicals
- Potential solutions
 - Eliminate pretreat if possible
 - Physically rinse every hole on a load
 - Use special rinse aids in the tank

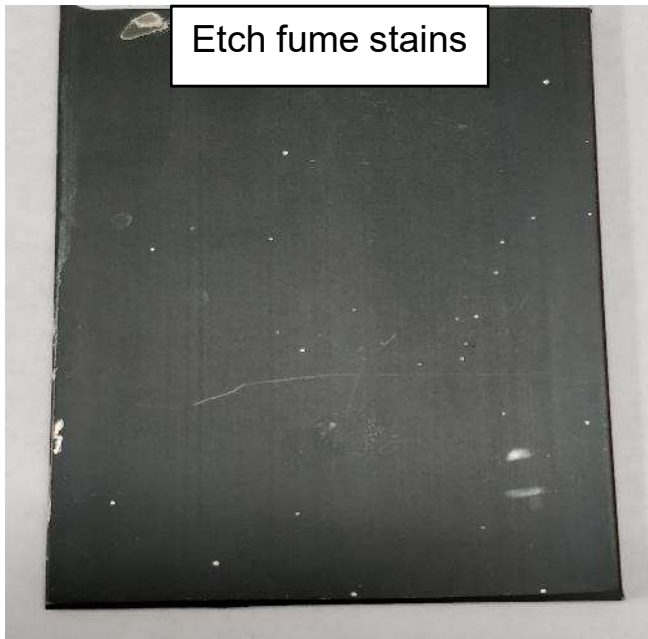
Contributing Factors: Pretreatment process

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Contributing Factors: Orientation & Production Line Airflow

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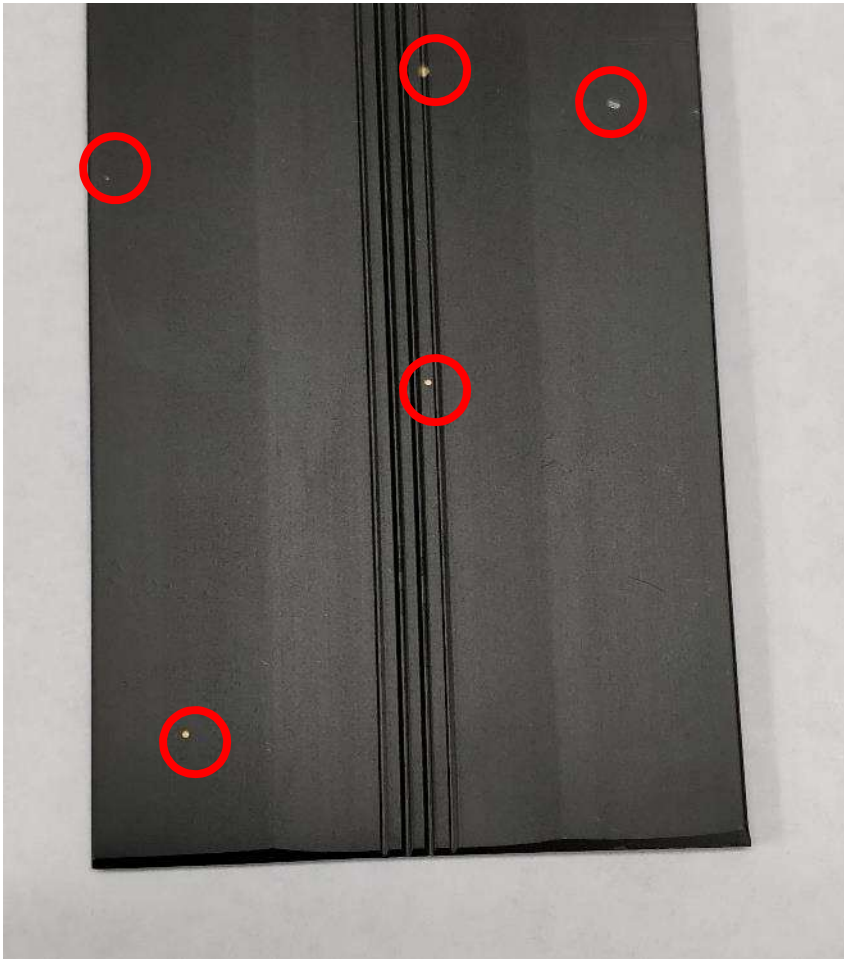
- Sources of spotting/staining
 - When an anodizing load comes out of the tank at the same time as an etch load caustic fumes can deposit on the anodized surface. Particularly if air flows from the etch tank
 - Etch fume staining occurs when etch dries on the part prior to rinsing and usually results in spotting on the top of the load instead of the bottom



- Potential solutions
 - Make sure the etch load and anodize load do not come out simultaneously
 - Reverse airflow to go from anodization tank to caustic tank by using industrial fans
 - Reduce time between etch and rinse or dwell time over etch tank
 - Check etch chemistry

Contributing Factors: Coating Passivation

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- Sources of spotting/staining
 - Extended immersion times in rinse water (greater than 10 minutes)
 - Rinse water contamination
 - Inefficient scheduling/planning
- Potential Solutions
 - Reduce immersion times in rinse tank
 - Make sure the parts are rinsed well through higher frequency immersions for shorter durations

Contributing Factors: Rinse Cycle



- Sources of spotting/staining
 - Insufficient rinsing after anodizing
 - Poor rinse water quality
 - Certain alloys are more susceptible to poor rinsing
- Potential Solutions
 - Improve rinse process using a spray rinse or higher frequency immersions
 - Use nitric rinse if possible

Contributing Factors: Rinse Cycle

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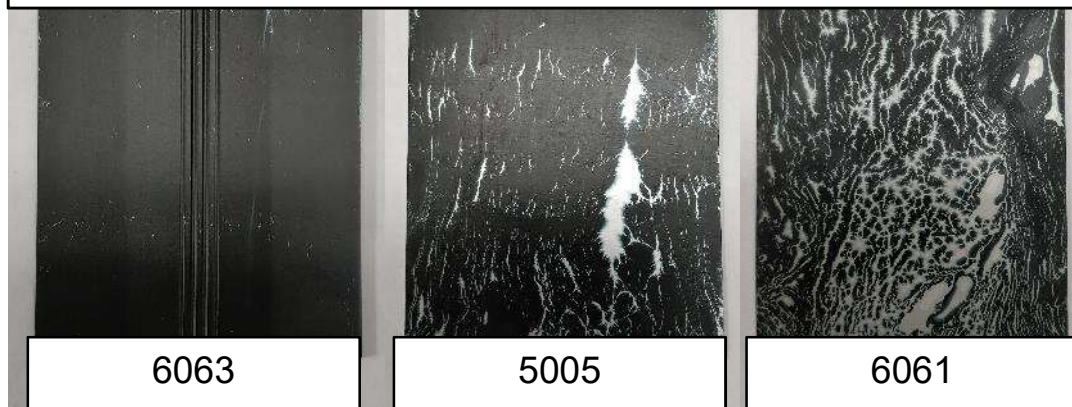
Alloy 6061



Alloy 5005

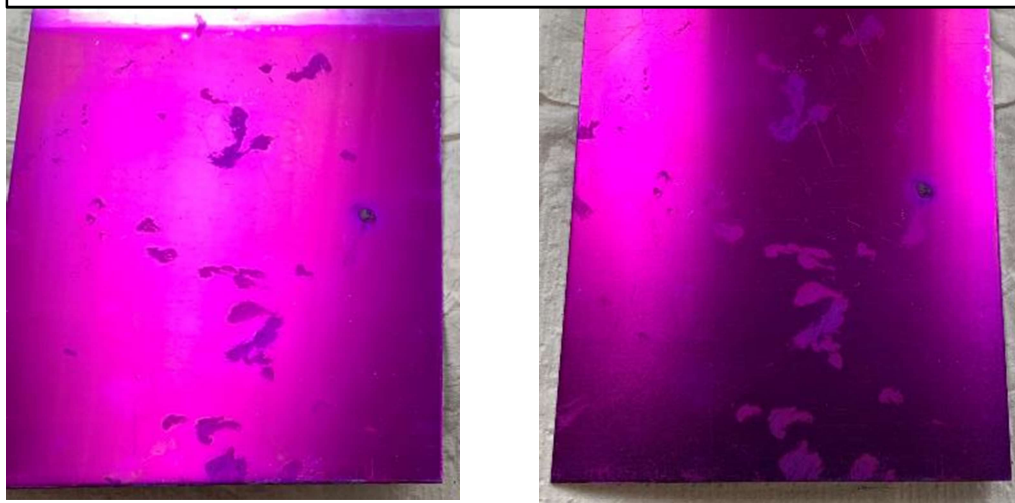
Contributing Factors: Dye Conditions

Improper dedusting agents

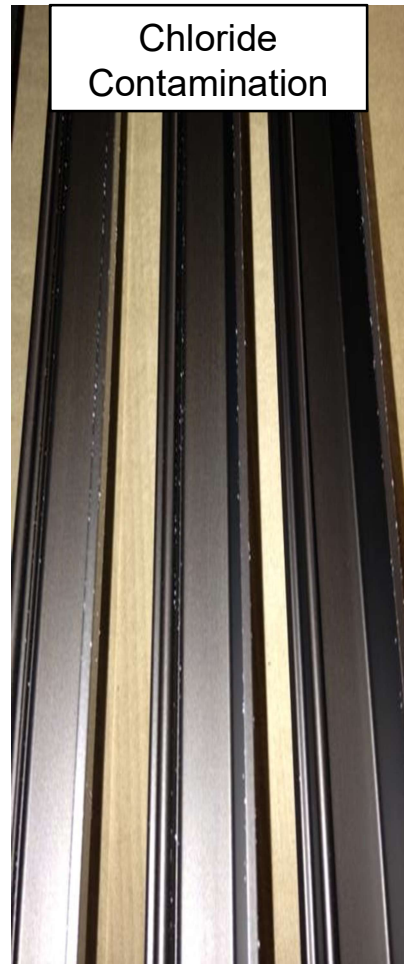


- Sources of spotting/staining
 - Use of improper dedusting agents in the dye.
 - Chlorides in dye
 - Chlorides in water
 - Algae contamination
- Potential Solutions
 - Make sure to purchase dye with proper dedusting agents
 - Use algaecides where possible
 - Contact supplier to determine if chlorides are present in dye
 - Adjust pH and temperature

Algae Contamination

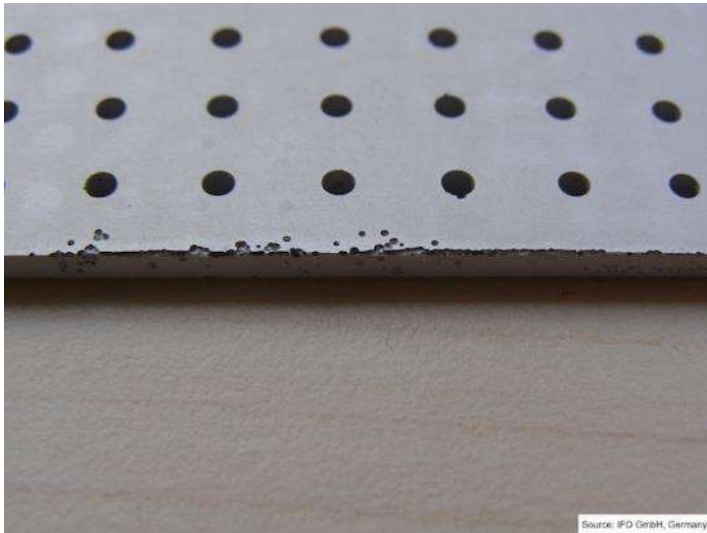


Contributing Factors: Sealing Process



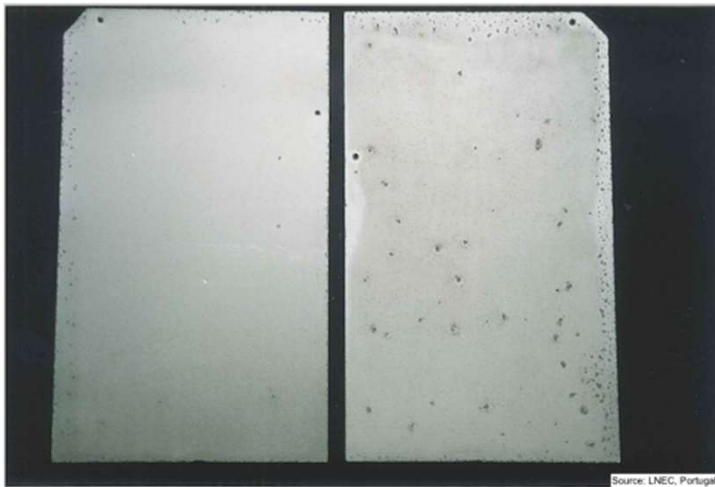
- Sources of spotting/staining
 - Chloride contamination
 - Precipitated Salts
 - Surfactant drying
 - Algae residue
- Potential solutions
 - Check water quality
 - Review temperature of seal chemistry
 - Check pH of seal

Surface Defect Sources: Corrosion



Galvanic Corrosion

- Galvanic Pitting occurs when dissimilar metals are electrically connected and the less noble of the metals corrodes. Particularly at sites where there are distinct breaks of the alloy constituents in the anodic coating.
- Galvanic activity can happen in several different process tanks and is random across a part which makes it difficult to identify.
- There are several factors that can cause Galvanic activity:
 - Alloy type (1100, 2000, 5083 and 7075).
 - Sharp machined edges
 - Chemistry contaminants
 - Extended immersion times
 - Titanium Tooling



Atmospheric and Other Corrosion

- Atmospheric or fall out corrosion can occur on incoming metal as a result of poor material handling
- Fall out corrosion typically occurs in specific areas of the part (outside surface) instead of the entire part
- There are several factors that contribute to fallout corrosion:
 - Contaminated cutting fluid (i.e. algae)
 - Moisture in part storage areas

Surface Defect Sources: Water Contaminants

Chloride
Contamination



Algae
Contamination



- Water contaminants can include the following:
 - Chlorides
 - Algae
- Chloride levels can vary significantly depending on the source of water therefore using DI water for all tanks after the pre treatment will significantly reduce the likelihood of chlorides
- Additionally chemistry should be checked for chlorides therefore contact your vendor to understand what chlorides if any could be in the following products:
 - Anodizing Additives
 - Post Anodize Rinse Additives
 - Dyes
 - Seals
- Algae is typically linked to dyes because many dyes are consumed by algae. Therefore dye tanks should be regularly dosed with algaecides particularly in summer months
- Furthermore when storing dyes for extended periods of time be sure to follow supplier guidelines on storage. In many cases liquid dyes are more susceptible to algae contamination.

Surface Defect Sources: Caustic Fume Staining



- Caustic fume staining occurs when parts are removed from the etch tank and are exposed to etch fumes for extended periods or when the etch dries on the part without immediate rinsing.
- There are several factors that can contribute to caustic fume staining/drying
 - Crane lift time
 - Ventilation/air flow around etch tank
 - Timing between etch and first rinse
 - Etch Chemistry

Surface Defect Sources: Coating Passivation



- Coating passivation occurs after a part is anodized and left in a rinse tank for an extended period of time (i.e. greater than 10 minutes) prior to dyeing or sealing.
- This extended immersion “passivates” the coating and essentially seals it to prevent any dye or seal to enter the coating.
- There are several factors that can contribute to passivation
 - Rinse tank availability
 - Scheduling process
 - Rinse water contaminants/temp

Surface Defect Sources: Bleed Out



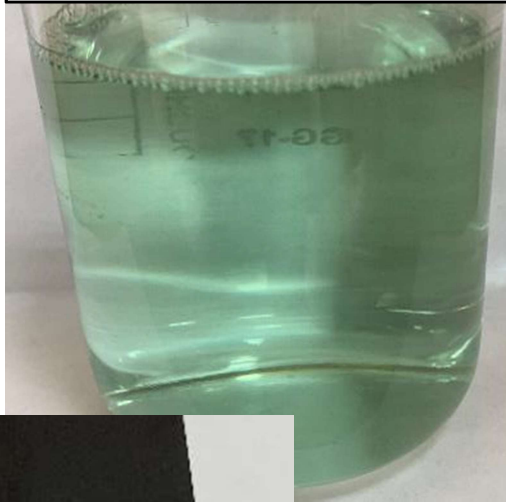
- Bleed out occurs when hot etch solutions cause threaded holes to expand and trap caustic which carries over to till the next hot process tank which is the dye tank
- When the etch “bleeds out” it creates white stains that do not take color
- There are several factors that can contribute to bleed out
 - Part orientation
 - Rinse cycle and frequency
 - Rinse aid chemistry

Surface Defect Sources: Precipitates / Solids Residue

Nickel Hydroxide
Precipitate



Normal Nickel Seal



- Precipitates and other solids residue on a part are most often seen during the final stages of the anodizing process such as dyeing or sealing
- There are several factors that can cause precipitates or other residue:
 - Nickel Hydroxide precipitate from seal
 - Surfactant drying from seal
 - Algae residue from dye or rinse tanks prior to seal
- Nickel Hydroxide formation this can be linked to a high pH (>6.0). When this happens the nickel hydroxide precipitate forms and cannot go back into solution

Methods to Mitigate Surface Defects

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Area of Focus	Cause	Potential Solution
<i>Alloy Type</i>	Galvanic Corrosion	- Manage such alloys by using aluminum tooling - Use Mg bars as sacrificial anodes
<i>Machining Quality / Part Profile</i>	- Fall Out Corrosion - Solution entrapment - Edge effect	- Check for corrosion patterns that are non random - Avoid sharp edges where possible - Increase etch times if possible or deploy magnesium bars - Work with customers to implement a slight radius - Reduce coating thickness if possible
<i>Pre-Treatment Process</i>	- Galvanic Corrosion - Bleed Out	- Eliminate etch if possible - Deploy physical rinsing techniques - Use special rinse aids - Avoid excessive immersion times in acidic rinses and process tanks
<i>Orientation / Production Line Airflow</i>	Caustic Fume Drying	- Install industrial fans in proper locations - Avoid simultaneous removal of anodizing and etch loads - Reduce the time loads are over the etch tank - Review etch chemistry conditions

Methods to Mitigate Surface Defects continued

Area of Focus	Cause	Potential Solution
<i>Anodizing Conditions</i>	• Galvanic Corrosion • Algae	• Reduce coating thickness wherever possible • Check for chlorides in anodizing tank & chemistry • Periodically kill algae using bleach in rinse tanks
<i>Coating Passivation</i>	• Excessive Rinse Times	• Reduce rinse immersion times to no more than 10 minutes by better scheduling
<i>Rinse Cycle</i>	• Chloride Contamination • Algae • Galvanic Corrosion	• Use spray rinses where possible • Check and control contaminants frequently • Increase rinsing cycle or incorporate a nitric rinse to remove sulfates • Use Mg bars as sacrificial anodes
<i>Dye Conditions</i>	• Chloride Contamination • Algae • Galvanic Corrosion • Dedusting agents	• Select dyes with low salt/chloride content • Select dyes with correct dedusting agents • Increase pH and reduce temperature and dye time if galvanic corrosions is an issue • Use Mg bars as sacrificial anodes
<i>Sealing Process</i>	• Chloride contamination • Surfactant drying • Precipitate salts	• Check and manage chloride levels in seal tank • Manage pH and temperature of seal chemistry closely

Questions

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Pinakin Patel

- Prior to starting Techevon, Pinakin served as Vice-President of Technology at Pioneer Metal Finishing Corporation. His emphasis was to introduce new products and processes that complement the business, helping Pioneer Metal Finishing harness more value from their existing customer base.
- Before joining Pioneer, Pinakin was the Industry Line Director of the Metals Business at the Clariant Corporation. Pinakin was awarded multiple patents on new methods of coloring and sealing technologies that used environmentally friendly chemistry. His industry-leading work on sealing became the basis of further work in the trade, resulting in more environmentally friendly products that are rapidly replacing older technologies. He has published papers in several magazines and has participated on various boards for discussions on technical issues.
- Pinakin has a Bachelor and Masters degree in Chemical Engineering. He began his career at Haviland Chemical Company in Grand Rapids, MI where he worked on chemistry for finishing aluminum. At Haviland, Pinakin developed a number of leading chemistries for anodizing and was a founding member of the Aluminum Anodizing Council.



Tej Patel

- Tej has over 10 years of consulting and industry experience specializing in the areas of supply chain, operations improvement and manufacturing. Prior to joining Techevon, he was a Vice President at AlixPartners, a leading global consulting firm. As part of the supply chain and operations practice, he led a number of cost savings and efficiency initiatives among a variety of manufacturers. Additionally, Tej has held positions at Baxter Inc., a Fortune 150 medical products manufacturer, and Archstone Consulting. While at Archstone Consulting, he led multiple client teams and projects in the areas of plant floor operations, LEAN manufacturing, and supply chain improvements across several industries.
- Tej holds an MBA from the Kellogg School of Management, with concentrations in Operations, Strategy, and Marketing in addition to a Bachelor's Degree in Biomedical Engineering from Northwestern University.

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